Junru Lu

Foundation Models / Efficient Attention / Agent Memory

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Education

Ph.D. in Computer Science — Nov. 2019 – June 2024
University of Warwick (Advisor: Prof. Yulan He)

M.S. in Applied Urban Science & Informatics — Sept. 2018 – Sept. 2019
New York University

Double Bachelor of Economics — Sept. 2015 – July 2018
Peking University

B.Eng. in Information Management and Information System — Sept. 2014 – July 2018
University of International Relations (Advisor: Prof. Binyang Li)

Career

Foundation Model Researcher, Tencent YouTu Lab — May 2023 – present
Core R&D member of the LLM and MLLM team — Shanghai, China

- **Data**: Built text and multimodal pre-training pipelines for quality filtering, deduplication, contamination detection and data synthesis. Text corpus: 20TB+; multimodal corpus: 5TB+.
- **Arch**: Made architecture decisions for Dense/MoE language and multimodal models across scales, including sparse attention and native multimodal modeling.
- **Infra**: Contributed to fused attention, routing and multimodal packing kernels, and co-designed the new LightningMoBA algorithm.
- **Train**: Explored distributed pre-training of 2B/4B/30B-A3B LLMs and 32B-A4B/120B-A7B/640B-A31B MoE MLLMs, with monitoring of downstream metrics, model health and training stability.
- **Eval**: Built automated inference validation and text, multimodal, long-context and agentic evaluation/reporting pipelines, and designed new benchmarks.

Patents: My patent portfolio covers SSA, ContextPilot, ElephantBench and LightningMoBA.

Selected Publications

- **[From-scratch 2B LLM] Youtu-LLM: Unlocking the Native Agentic Potential for Lightweight Large Language Models**
  Junru Lu, Jiarui Qin, et al. (2025). arXiv:2512.24618.
- **[From-scratch 4B MLLM] Youtu-VL: Unleashing Visual Potential via Unified Vision-Language Supervision**
  Zhixiang Wei, Yi Li, ..., **Junru Lu**, et al. (2026). arXiv:2601.19798.
- **Toward Native Multimodal Modeling: A Roadmap**
  Siyu An, **Junru Lu**, et al. (2026). EMNLP 2026.
- **ContextPilot: Teaching Agents for Proactive Context Management via Fine-grained RL**
  Zhuoshi Pan, Qizhi Pei, **Junru Lu**, et al. (2026). EMNLP 2026 (Main).
- **SSA: Sparse Sparse Attention by Aligning Full and Sparse Attention Outputs in Feature Space**
  Zhenyi Shen, **Junru Lu**, et al. (2026). ICML 2026.
- **OptiScene: LLM-driven Indoor Scene Layout Generation via Scaled Human-aligned Data Synthesis and Multi-Stage Preference Optimization**
  Yixuan Yang, Zhen Luo, ..., **Junru Lu**, et al. (2025). NeurIPS 2025.
- **Two Heads Are Better Than One: Dual-Model Verbal Reflection at Inference-Time**
  Jiazheng Li, Yuxiang Zhou, **Junru Lu**, et al. (2025). EMNLP 2025.
- **RoleMRC: A Fine-Grained Composite Benchmark for Role-Playing and Instruction-Following**
  Junru Lu, Jiazheng Li, et al. (2025). ACL 2025 (Findings).
- **Tell Me What You Don't Know: Enhancing Refusal Capabilities of Role-Playing Agents via Representation Space Analysis and Editing**
  Wenhao Liu, Siyu An, **Junru Lu**, et al. (2025). ACL 2025 (Findings).
- **Eliminating Biased Length Reliance of Direct Preference Optimization via Down-Sampled KL Divergence**
  Junru Lu, Jiazheng Li, et al. (2024). EMNLP 2024.
- **FIPO: Free-form Instruction-oriented Prompt Optimization with Preference Dataset and Modular Fine-tuning Schema**
  Junru Lu, Siyu An, et al. (2025). COLING 2025.
- **Advancing Video Anomaly Detection: A Bi-Directional Hybrid Framework for Enhanced Single- and Multi-Task Approaches**
  Guodong Shen, Yuqi Ouyang, **Junru Lu**, et al. (2024). IEEE Transactions on Image Processing.
- **NapSS: Paragraph-level Medical Text Simplification via Narrative Prompting and Sentence-matching Summarization**
  Junru Lu, Jiazheng Li, et al. (2023). EACL 2023 (Findings).
- **Event-Centric Question Answering via Contrastive Learning and Invertible Event Transformation**
  Junru Lu, Xingwei Tan, et al. (2022). EMNLP 2022 (Findings).
- **CHIME: Cross-passage Hierarchical Memory Network for Generative Review Question Answering**
  Junru Lu, Gabriele Pergola, et al. (2020). COLING 2020.